

ISHAN BIJUKUCHHAY

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EDUCATION

Queen's University - Bachelor of Applied Science, Computer Engineering September 2025 - April 2030

PROJECT EXPERIENCE

Bare-Metal ARM Firmware (STM32F401RE, Cortex-M4) August 2026 - Present
Personal Project

- Wrote a bare-metal driver stack in C with no HAL or generated code: custom linker script, vector table, and C runtime startup (`.data` flash-to-RAM copy, `.bss` zeroing) derived from the reference manual.
- Implemented register-level GPIO, UART (interrupt-driven RX with a lock-free ring buffer), I2C with a BME280 sensor and fixed-point compensation, a SysTick timebase, and DMA-driven UART transmit.
- Brought the clock tree from the 16 MHz HSI to the part's 84 MHz maximum through the PLL, re-deriving and re-measuring every clock-dependent constant (baud, I2C timing, SysTick reload) on hardware.
- Verified at every level: GDB register reads, disassembly of the flashed binary, and logic-analyzer protocol decode; diagnosed I2C sequencing and race-condition bugs from peripheral state registers alone.

HTTP Caching Reverse Proxy July - August 2026
Personal Project

- Built an HTTP caching reverse proxy in C on a Raspberry Pi 5 (hash table with separate chaining, $O(1)$ lookup; doubly-linked LRU list, $O(1)$ eviction).
- Benchmarked on hardware with `wrk`: $\sim 46k$ req/s on cache hits vs $\sim 1.6k$ on origin fetches ($\sim 29\times$ throughput).
- Profiled four concurrency models (single-threaded, thread-per-connection, thread pool, epoll) and eviction policies (LRU held 75% hit rate vs LFU's 9.5% under drifting load).

EcoTrack - Carbon Footprint Tracker 2026
Team Lead, APSC 103

- Led a team of 5 to ship a full-stack web app (React, Flask) converting everyday choices into CO₂e estimates; managed deadlines and client communication through to a working prototype.

LEADERSHIP EXPERIENCE

Software Project Lead August 2026 - Present
Queen's VEX U Robotics

- Lead the software subteam building controls and autonomy for a real-time robotic drivetrain, owning technical direction and mentoring members.
- Wrote C++ for sensor-driven autonomous behaviour on an intake-and-conveyor scoring robot; integrated sensors and debugged signal paths between control electronics and actuators.
- Coordinate with the mechanical subteam on models, constraints, and performance requirements, and run planning and check-ins to keep the team aligned with club directors.

WORK EXPERIENCE

Lifeguard/Instructor June 2025 - September 2025
City of Kitchener - Aquatics Division

- Planned, delivered, and adapted over 20 swim programs for 100+ participants across age groups.

SKILLS AND INTERESTS

Languages	C, C++, Rust, Python
Embedded	Bare-metal ARM Cortex-M4, UART, I2C, DMA, interrupts & NVIC, SysTick, linker scripts
Systems	Data Structures & Algorithms, Multithreading, Sockets, Linux
Tools	Git, GDB, Make, GCC/arm-none-eabi, OpenOCD, logic analyzer (sigrok)
Hardware	PCB Design (Proteus), Fusion 360